Industrial Internship Report on

”AI-Powered Full-Stack Grocery Delivery Application”

Prepared by

Ujwala Uday Naik

Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was "Grocery Delivery Application," a Java Full Stack web-based application developed to simplify online grocery shopping and order management. The system allows customers to register, browse and search grocery products, manage their shopping cart, select delivery slots, choose a payment method and place orders seamlessly. An administrative dashboard enables efficient management of products, categories, inventory, customers and order processing. The application was built using Java, JSP, Servlets, JDBC, MySQL, HTML, CSS, Bootstrap and JavaScript, ensuring a responsive, secure and user-friendly platform for both customers and administrators. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

This internship provided me with an excellent opportunity to gain practical knowledge and hands-on experience in Java Full Stack Web Development. Over the course of six weeks, I learned modern web development concepts, software engineering practices, and the complete lifecycle of designing, developing, testing, and deploying a real-world web application. The internship strengthened my technical knowledge while improving my analytical thinking, problem-solving abilities, teamwork, and project management skills.

As part of the internship, I developed Grocery Delivery Application, a Java Full Stack web application designed to simplify online grocery shopping and order management. The application enables customers to register, browse grocery products, search items by category, add products to their shopping cart, select delivery slots, choose a payment method, and place orders conveniently. It also provides an Admin Dashboard that allows administrators to manage products, categories, inventory, customers, and orders efficiently. The application was developed using Java, JSP, Servlets, JDBC, MySQL, HTML, CSS, Bootstrap, and JavaScript, providing a responsive, secure, and user-friendly online grocery shopping experience.

The internship organized by Upskill Campus (USC) in collaboration with UniConverge Technologies (UCT) provided a structured learning environment with comprehensive learning resources, project guidance, and weekly assignments. The program was

systematically planned, beginning with the fundamentals of Java programming and full-stack web development, followed by database design, user interface development, backend implementation, testing, debugging, and project documentation. This step-by-step approach enabled me to successfully complete the project while continuously enhancing my technical and practical skills.

Throughout the internship, I gained valuable experience in Java, JSP, Servlets, JDBC, MySQL, HTML, CSS, Bootstrap, JavaScript, database connectivity, session management, CRUD operations, and responsive web application development. I also learned the importance of writing clean and maintainable code, designing efficient database structures, debugging applications, implementing user authentication, and developing scalable web-based solutions. These experiences significantly improved my confidence in building full-stack applications and understanding industry-standard software development practices.

I sincerely express my gratitude to Upskill Campus (USC) and UniConverge Technologies (UCT) for providing this valuable internship opportunity and an excellent platform for learning. I would also like to thank my mentors, trainers, project coordinators, and everyone who supported and guided me throughout the internship. Their encouragement, technical guidance, and constructive feedback played a vital role in the successful completion of my project.

Finally, I encourage my juniors and peers to make the best use of internship opportunities that provide practical exposure to real-world software development. Stay committed to learning, practice consistently, work on real-time projects, and continuously improve your technical skills. The practical knowledge and experience gained through such internships are invaluable in building a successful career in the software industry.

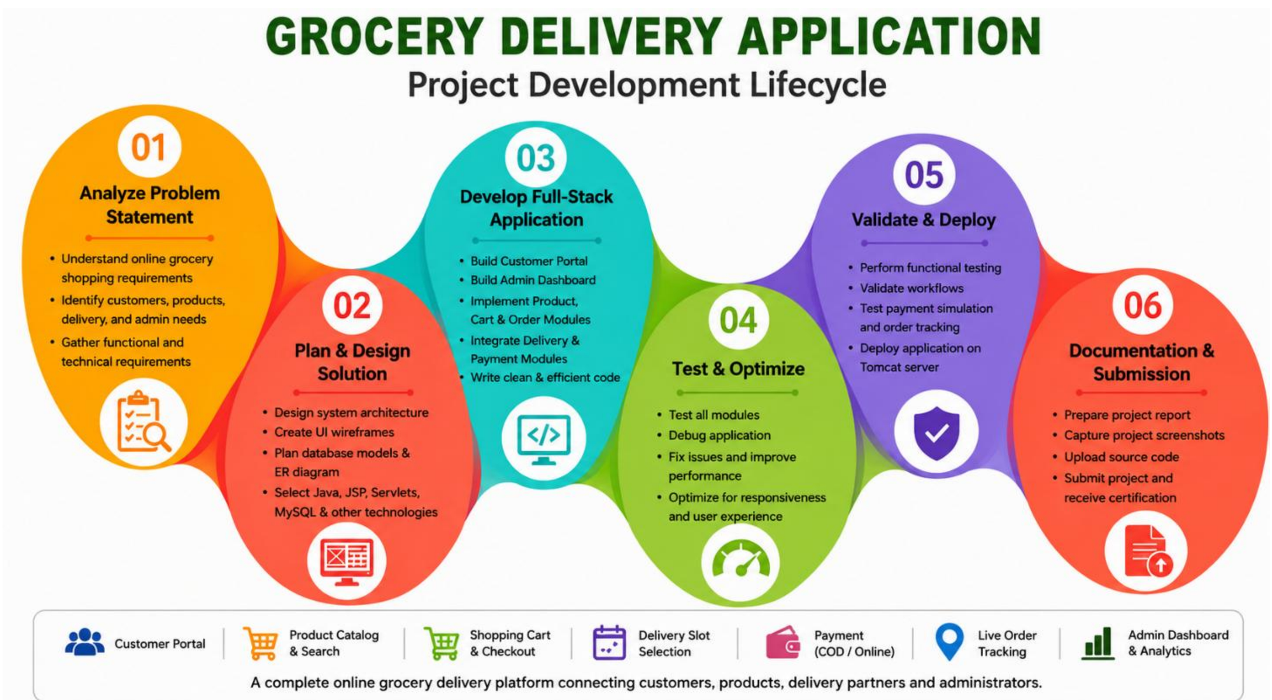


Fig 1.1 : Grocery Delivery Application Project Development Lifecycle

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

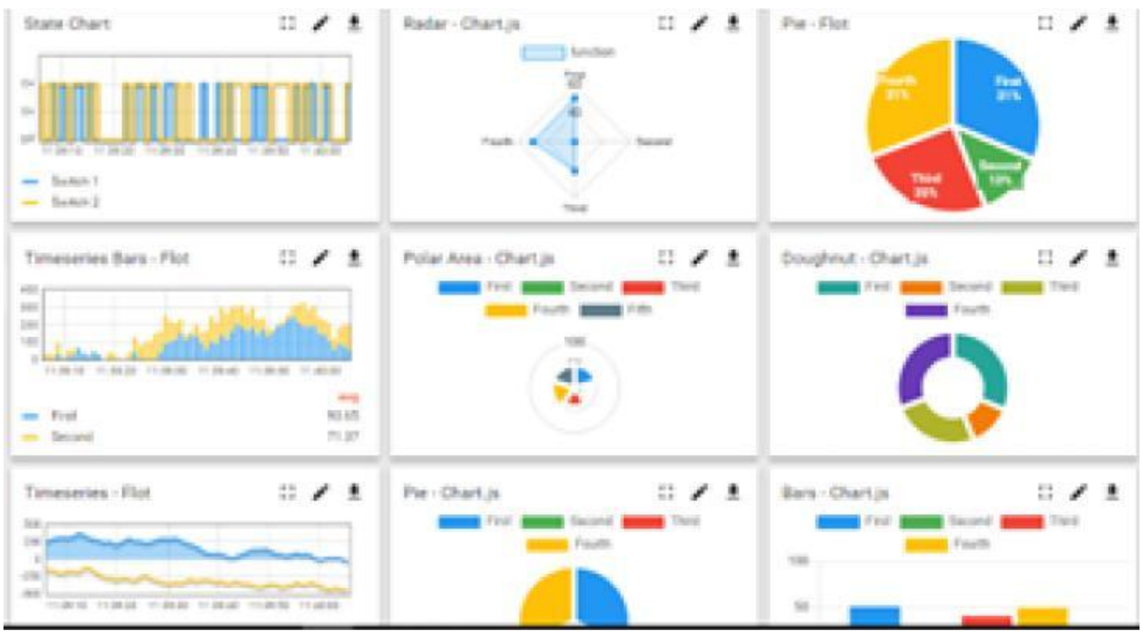
i. UCT IoT Platform ( )

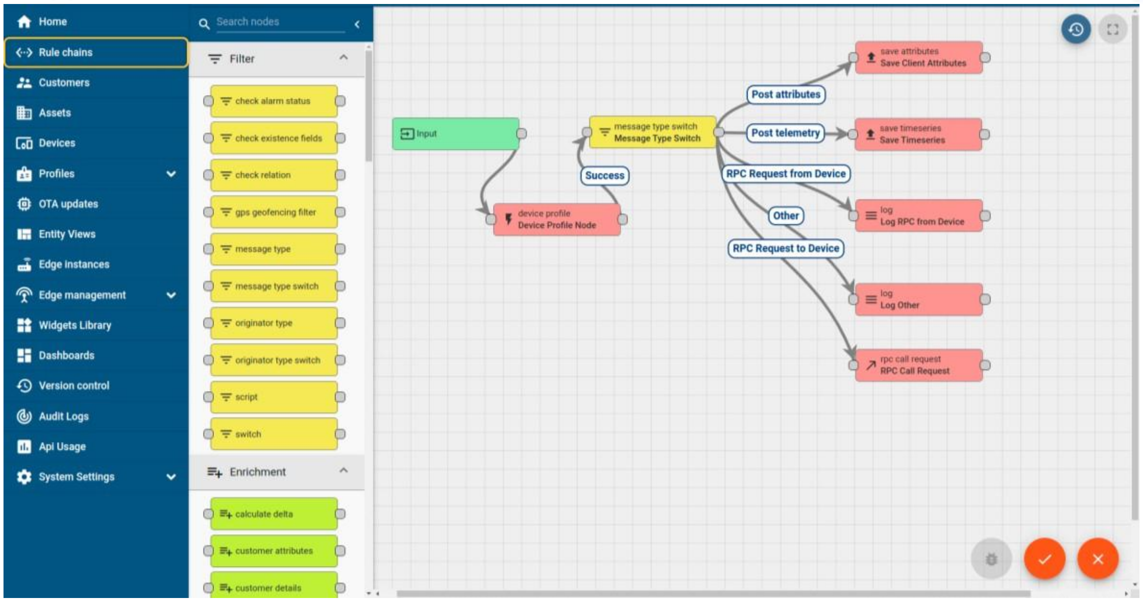
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
  - Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.


Demo Org
Plant Name
Demo
Plant Name
0%
Plant utilization
55%
Average OEE
4
Capacity
0
Utilized
0
Planned Production
0
Plant Operation
Iso Front Back Right Left Top Bottom
CNC_S7_81
CNC_S7_82
CNC_S7_84
Direct Email Union
Utilization
58%
Assist OEE
522 days
Spindle RUL
53%
521 days
Machine | Operator | Work Order ID | Job ID | Job Performance | Job Progress | Start Time | End Time | Output | Planned | Actual | Rejection | Time (mins) | Setup | Pred | Downtime | Idle | Job Status | End Customer
CNC_S7_81 | Operator 1 | WO0405200001 | 4168 | 58% | 10:30 AM | 55 | 41 | 0 | 80 | 215 | 0 | 45 | In Progress | i
CNC_S7_81 | Operator 1 | WO0405200001 | 4168 | 58% | 10:30 AM | 55 | 41 | 0 | 80 | 215 | 0 | 45 | In Progress | i


3D View
12:30
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Stopped
Workplace Type
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Availability
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Monitor
42.23%
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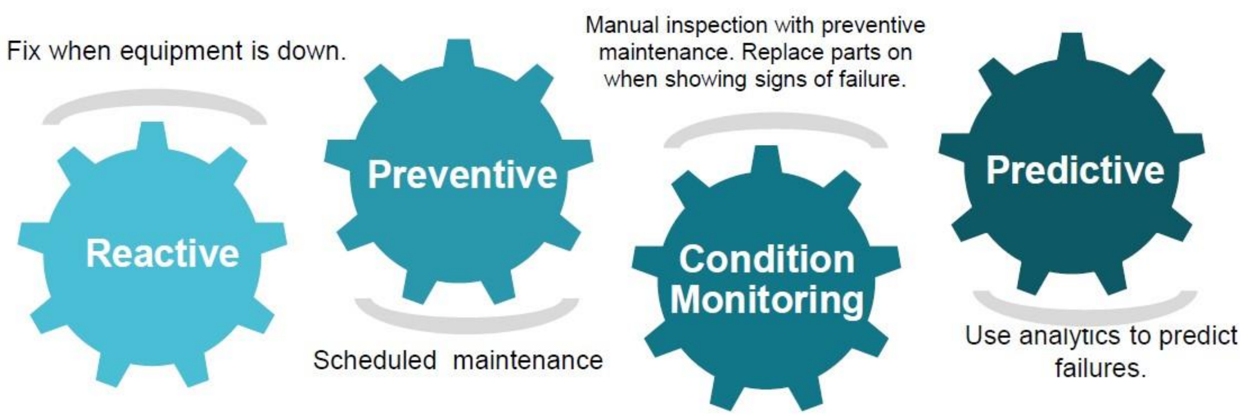
99.21%
24.53%
62.35%
99.21%
24.53%
62.35%
Availability
Performance
Quality
OEE

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc. iv. Predictive Maintenance

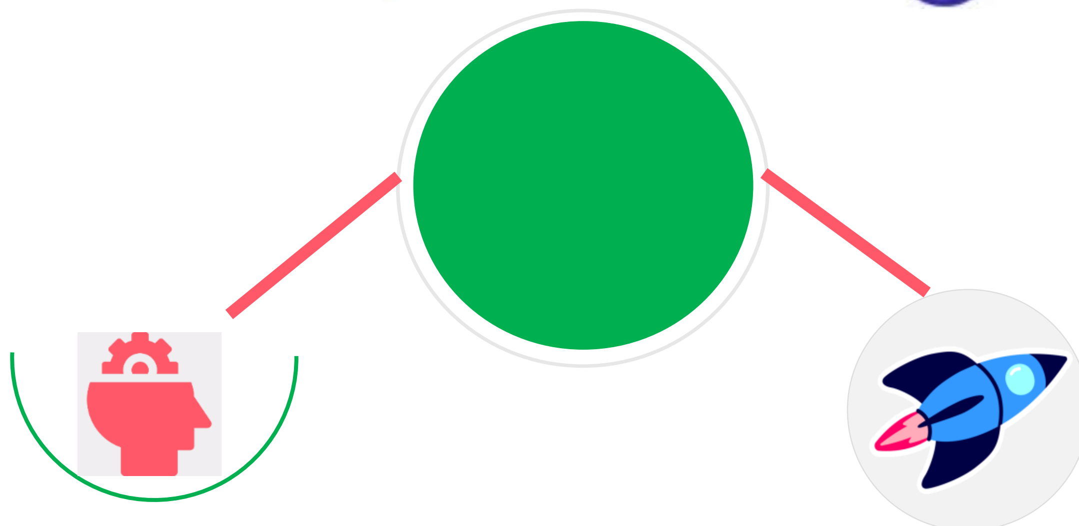
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

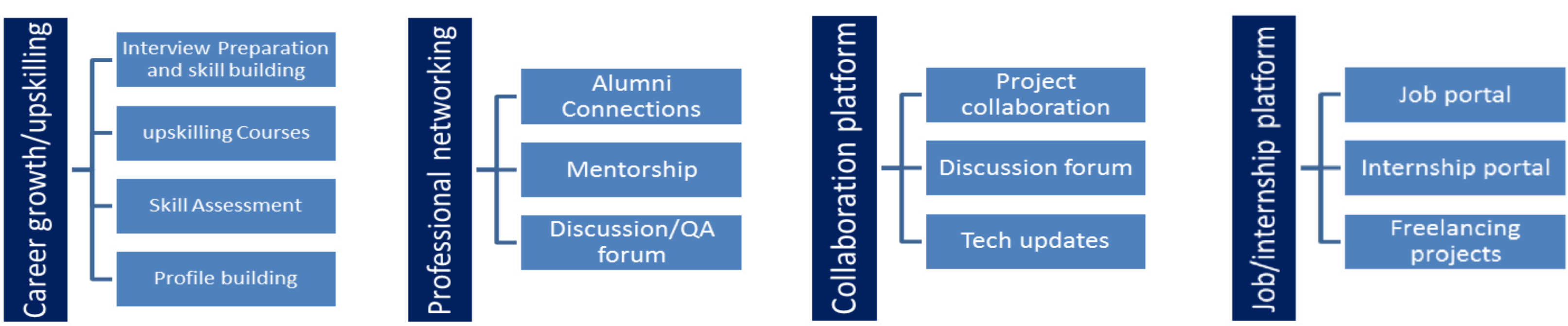
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship Program

The objectives of this internship program were to:

- Gain practical experience in Java Full-Stack Web Development.
- Understand the Software Development Life Cycle (SDLC).
- Develop a real-world Grocery Delivery Application using Java technologies.
- Learn JSP, Servlets, JDBC, MySQL, HTML, CSS, Bootstrap, and JavaScript.
- Implement user authentication, product management, shopping cart, and order processing.
- Understand database design and frontend-backend integration.
- Improve programming, debugging, testing, and problem-solving skills.
- Enhance teamwork, communication, and project management skills.

2.5 Reference

[1] React Official Documentation – https://react.dev

[2] Express.js Documentation – https://expressjs.com

[3] Node.js Documentation – https://nodejs.org

[4] Google Gemini AI Documentation – https://ai.google.dev

[5] Tailwind CSS Documentation – https://tailwindcss.com

2.6 Glossary

Terms	Acronym
Java Server Pages	JSP
Java Database Connectivity	JDBC
HyperText Markup Language	HTML
Cascading Style Sheets	CSS
JavaScript	JS
User Interface	UI
Hypertext Transfer Protocol	HTTP
Structured Query Language	SQL
Create, Read, Update, Delete	CRUD
Full Stack Development	FSD
Database Management System	DBMS
Cash on Delivery	COD
Java Development Kit	JDK
Apache Tomcat	Tomcat
Model View Controller	MVC

3 Problem Statement

The traditional grocery shopping process often requires customers to visit physical stores, spend considerable time searching for products, wait in long billing queues, and carry groceries back home. Customers may also face difficulties in checking product availability, comparing prices, selecting convenient delivery timings, and managing their grocery purchases efficiently. Similarly, store administrators face challenges in maintaining inventory, updating product information, processing customer orders, and managing deliveries effectively.

The objective of this project is to develop a Java Full Stack Grocery Delivery Application that provides a convenient and efficient online platform for grocery shopping. The system allows customers to register, browse grocery products, search products by category, view product details, add items to the shopping cart, select preferred delivery slots, choose payment methods, and place orders seamlessly. The application also enables administrators to manage product categories, update inventory, process customer orders, and monitor order status through a dedicated admin dashboard.

The application focuses on improving the overall shopping experience by providing features such as product search, category-wise browsing, shopping cart management, delivery slot selection, secure user authentication, online or cash-on-delivery payment options, order history, and efficient order management. The

responsive and user-friendly interface ensures that customers can conveniently purchase groceries anytime and anywhere.

The proposed solution, Grocery Delivery Application, aims to provide a simple, reliable, and scalable online grocery shopping platform that connects customers and administrators through a centralized system. By digitizing the grocery ordering process, the application improves shopping convenience, reduces manual effort, enhances inventory management, and delivers a seamless experience for both customers and store administrators.

4. Existing and Proposed Solution

4.1 Existing Solutions

Nowadays, many supermarkets and grocery stores still rely on traditional shopping methods where customers visit physical stores to purchase grocery items. Although several online grocery platforms are available, many small and medium grocery businesses do not have a dedicated web-based system for managing products, customer orders, inventory, and deliveries efficiently. Customers often spend considerable time searching for products, waiting in billing queues, and carrying groceries themselves.

Some existing grocery shopping systems also have certain limitations:

- Limited product search and category filtering.
- Lack of flexible delivery slot selection.
- Manual inventory management.
- Limited order tracking and order history.
- Separate systems for customer and administrator management.
- Less convenient shopping experience for customers.
- Difficulty in managing products and customer orders efficiently.
- High dependency on manual record keeping.

4.2 Proposed Solution

The proposed solution is Grocery Delivery Application, a Java Full Stack web application developed to provide a simple, secure, and user-friendly online grocery shopping platform. The application allows customers to purchase grocery products from anywhere while enabling administrators to efficiently manage products, categories, inventory, and customer orders.

The system provides two role-based portals:

1. Customer Portal

- Register and Login securely.
- Browse grocery products.

- Search products by name.
- View products category-wise.
- View product details including price and available quantity.
- Add products to the shopping cart.
- Update or remove cart items.
- Select a convenient delivery slot.
- Choose a payment method (Cash on Delivery or Online Payment Demo).
- Place grocery orders.
- View previous order history.
- Logout securely.

2. Admin Portal

- Secure Admin Login.
- Add new grocery products.
- Update product information.
- Delete products.
- Manage product categories.
- Manage inventory and product quantity.
- View registered customers.
- View customer orders.
- Update order status (Pending, Confirmed, Delivered).
- Monitor overall system activities.

4.3 Value Addition

The proposed Grocery Delivery Application offers several advantages over traditional grocery shopping systems.

- Simple and User-Friendly Interface: Designed with HTML, CSS, Bootstrap, and JavaScript to provide an easy shopping experience.
- Secure User Authentication: Allows secure registration, login, and session management using Java Servlets.
- Efficient Product Management: Administrators can easily add, update, delete, and organize grocery products and categories.
- Shopping Cart Management: Customers can add multiple products, modify quantities, and review their cart before placing an order.
- Category-Based Product Browsing: Products are organized into categories such as Vegetables, Fruits, Dairy, Bakery, Snacks, and Beverages for easy navigation.
- Flexible Delivery Slot Selection: Customers can choose a suitable delivery time while placing an order.
- Multiple Payment Options: Supports Cash on Delivery and Online Payment (Demo) for convenient checkout.
- Order Management System: Customers can view their order history, while administrators can manage and update order status efficiently.
- Inventory Management: Product quantity is automatically maintained, helping administrators monitor available stock.
- Responsive Web Application: The application works efficiently across desktops, laptops, tablets, and mobile devices.
- Centralized Administration: A dedicated admin dashboard enables efficient management of products, categories, customers, inventory, and orders from a single platform.

4.4 Codesubmission(Githublink)

https://github.com/ujwalanaik/upskillcampus

4.5 Reportsubmission(Githublink):

5. Proposed Design / Model

5.1 System Design Overview

The proposed system, Grocery Delivery Application, is a Java Full Stack web application designed to provide a convenient online platform for purchasing grocery products. The system connects customers and administrators through a centralized digital platform. Customers can browse and search for grocery items, manage their shopping cart, select delivery slots, choose payment methods, and place orders. Administrators can manage products, categories, inventory, customers, and orders through a dedicated admin portal.

The application follows a role-based architecture with two main user roles: Customer and Admin. The frontend is developed using HTML, CSS, Bootstrap, JavaScript, and JSP, while the backend uses Java Servlets and JDBC for application logic and database communication. MySQL is used to store user, product, cart, and order information.

5.1.1 Design Flow of the Proposed System

Step 1: User Registration and Authentication

The system begins with customer registration and authentication.

- Customer: Can register, log in, browse products, manage the cart, place orders, and view order history.
- Admin: Can log in through the admin portal to manage products, categories, customers, inventory, and orders.

User details are securely stored in the MySQL database, and session management is used to maintain authenticated users.

Step 2: Product and Category Management

The administrator manages the grocery products and categories through the admin dashboard.

The admin can:

- Add new grocery products.
- Update product information.
- Delete products.
- Add and manage product categories.
- Update product prices.
- Update available product quantities.
- Manage product availability.

Product information such as product name, description, price, quantity, category, and image is stored in the MySQL database.

Step 3: Customer Grocery Browsing and Selection

Customers can browse and select grocery products from the application.

Customers can:

- View all available grocery products.
- Browse products category-wise.
- Search for products by name.
- View product details.
- Check product price and available quantity.
- Select the required quantity.
- Add products to the shopping cart.

Example categories include:

- Vegetables
- Fruits
- Dairy
- Bakery
- Beverages
- Snacks
- Rice
- Cooking Oil
- Personal Care
- Household

Step 4: Cart and Order Processing

The shopping cart manages all products selected by the customer.

The cart calculates:

- Product quantity.

- Individual item price.
- Item subtotal.
- Total order amount.

Customers can:

- Add products to the cart.
- Increase or decrease quantity.
- Remove products.
- Review the total amount.

After reviewing the cart, the customer proceeds to checkout.

Step 5: Delivery Slot and Payment Selection

During checkout, customers provide or select their delivery information and choose a preferred delivery slot.

Available delivery slots include:

- 9 AM – 11 AM
- 11 AM – 1 PM
- 2 PM – 4 PM
- 5 PM – 7 PM

The customer can choose a payment method:

- Cash on Delivery (COD)
- Online Payment – Demo

After selecting the required options, the customer confirms the order.

Step 6: Order Confirmation and Management

Once the customer places an order, the order details are stored in the database.

The order contains information such as:

- Order ID
- Customer details
- Ordered products

- Product quantities
- Total amount
- Delivery slot
- Payment method
- Order date
- Order status

The order status can be updated by the administrator according to the processing stage:

PENDING → CONFIRMED → OUT FOR DELIVERY → DELIVERED

Customers can view their order status and previous orders through the Order History section.

Step 7: Admin Management and Inventory Control

The administrator manages the overall grocery delivery system through the admin dashboard.

The admin can:

- View registered customers.
- Manage products.
- Manage categories.
- Monitor product quantities.
- View customer orders.
- Update order status.
- Maintain product availability.
- Manage inventory information.

This centralized management system reduces manual work and helps maintain accurate product and order records.

Step 8: Database and Application Communication

The application uses JDBC to establish communication between the Java backend and MySQL database.

The overall communication flow is:

Customer/Admin → JSP Interface → Java Servlet → DAO/JDBC → MySQL Database

The database stores and retrieves information related to:

- Users
- Admins
- Products
- Categories
- Cart
- Orders
- Order Items

This architecture provides a structured approach for managing application data and business operations.

Step 9: Final Order Completion

After the order is delivered, the administrator can update the order status to Delivered. Customers can then view the completed order in their order history.

The complete workflow of the proposed Grocery Delivery Application is:

REGISTER/LOGIN → BROWSE PRODUCTS → SEARCH/FILTER → ADD TO CART → CHECKOUT → SELECT DELIVERY SLOT → SELECT PAYMENT METHOD → PLACE ORDER → ORDER CONFIRMATION → ORDER STATUS → DELIVERED → ORDER HISTORY

This design provides a simple, efficient, and user-friendly grocery shopping experience while allowing administrators to manage the application's products, inventory, customers, and orders from a centralized system.

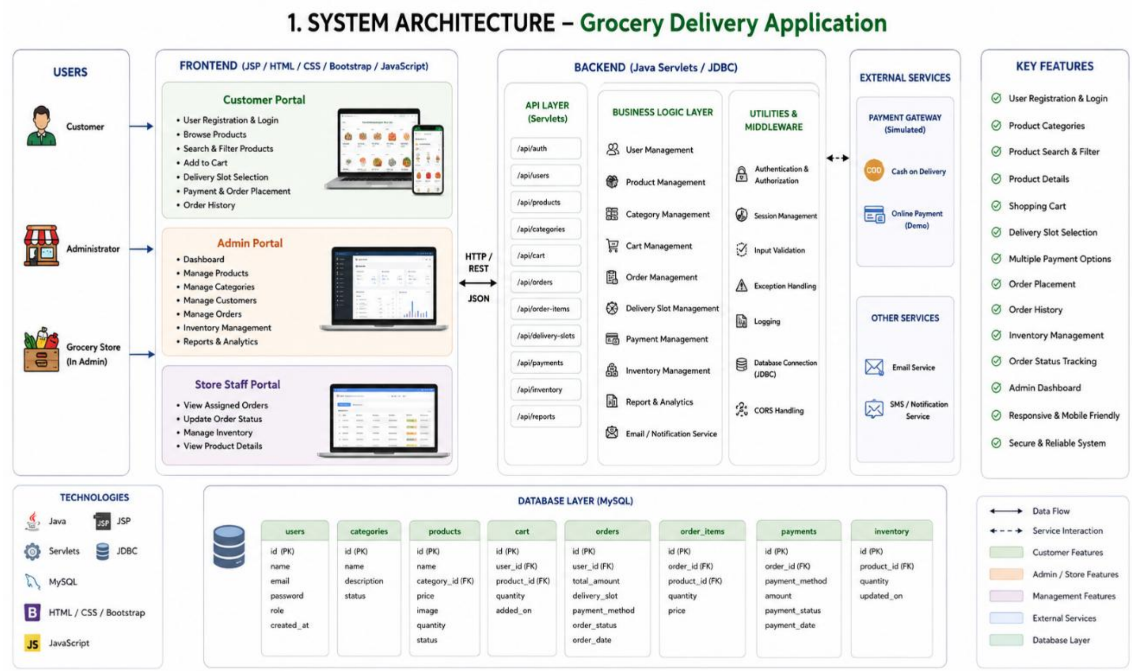


Figure 1: SYSTEM ARCHITECTURE DIAGRAM OF THE SYSTEM

5.2 High Level Diagram

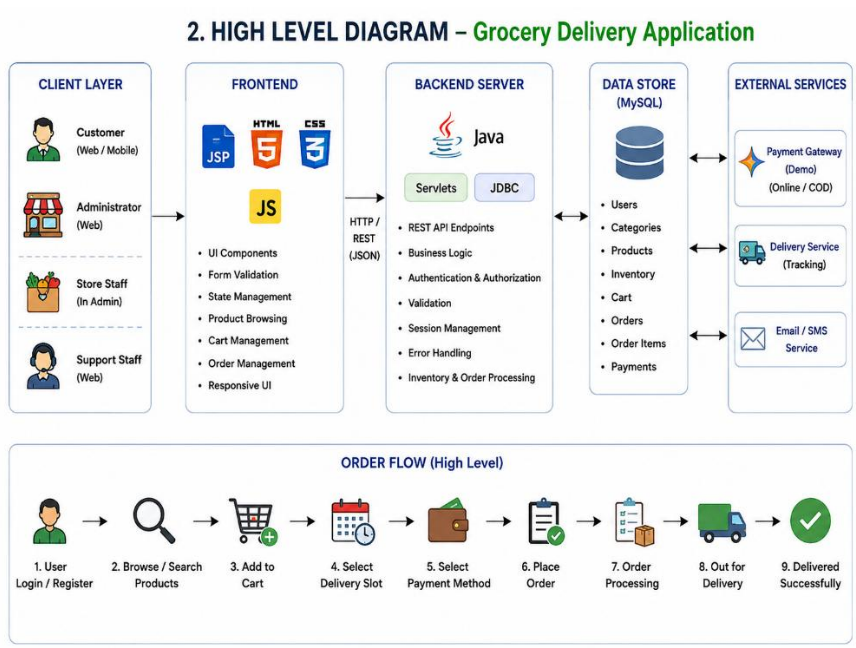


Figure 2: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.3 Low Level Diagram

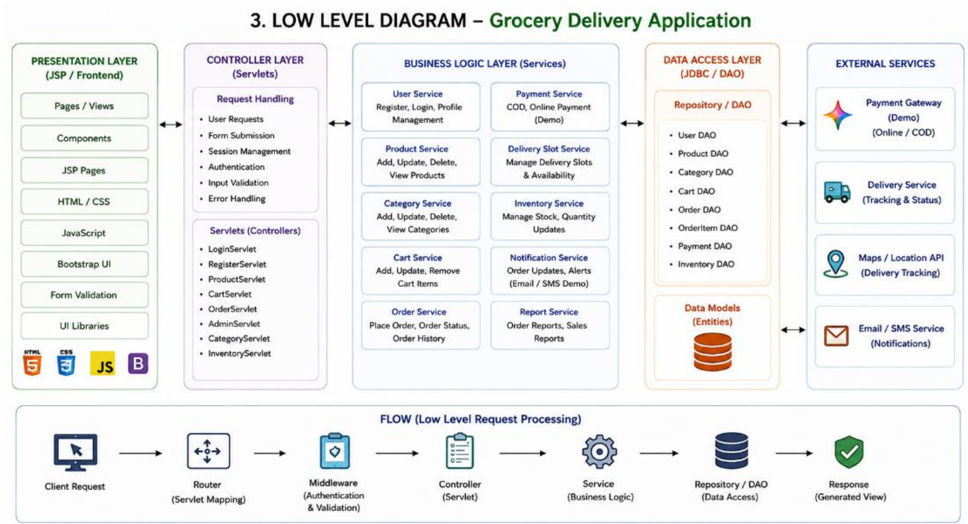


Figure 3: LOW LEVEL DIAGRAM OF THE SYSTEM

5.4 Interfaces

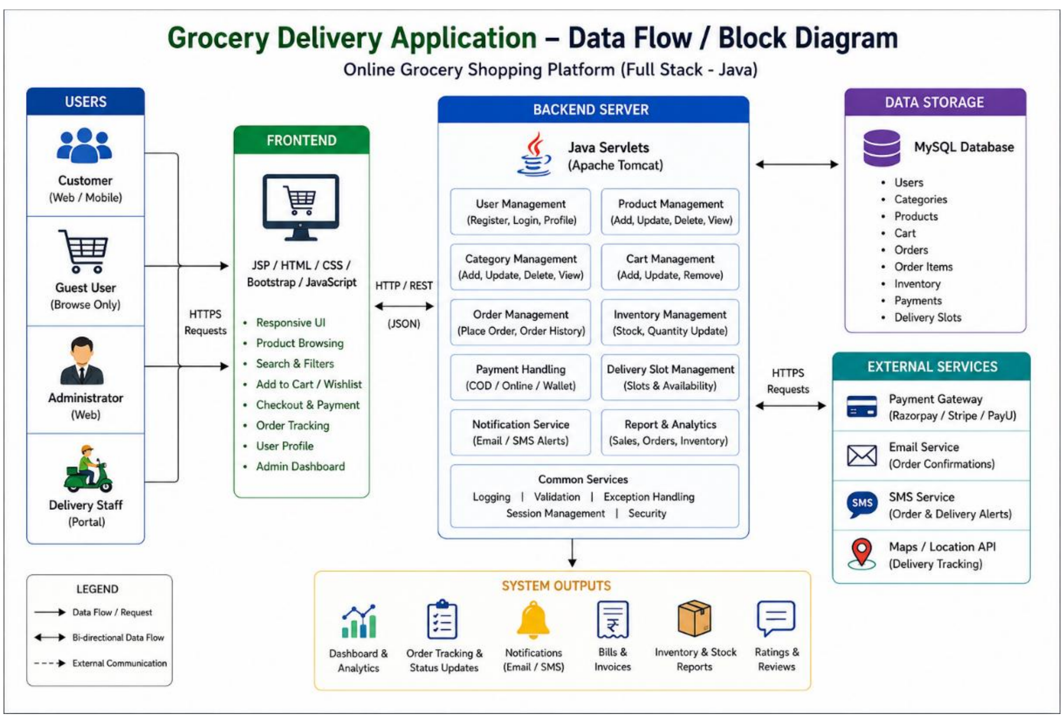


Figure 4: DATA FLOW DIAGRAM OF THE SYSTEM

6. Performance Test

The performance testing of the Grocery Delivery Application was conducted to evaluate the responsiveness, reliability, and efficiency of the application under normal operating conditions. The application was tested for page loading, database response time, product search, shopping cart operations, order processing, login authentication, and overall system performance.

The testing focused on ensuring that customers can browse products, manage their carts, select delivery slots, and place orders smoothly, while administrators can efficiently manage products, inventory, customers, and orders through the admin dashboard. The application was also checked for proper database connectivity and reliable data retrieval using Java Servlets, JDBC, and MySQL.

The performance testing confirmed that the major modules of the application function correctly and provide a smooth user experience under normal usage conditions.

6.1 Test Plan/ Test Cases

Test ID	Test Case	Expected Result	Status
PT-01	Load Home Page	Home page loads correctly within the expected time	Pass
PT-02	User Registration	New customer account is created successfully	Pass
PT-03	User Login	Valid user credentials allow successful login	Pass
PT-04	Search Products	Relevant grocery products are displayed correctly	Pass
PT-05	View Product Details	Product name, price, quantity, and category are displayed correctly	Pass
PT-06	Add Product to Cart	Selected product is added to the shopping cart successfully	Pass
PT-07	Update Cart Quantity	Cart quantity and total amount are updated correctly	Pass
PT-08	Checkout and Delivery Slot	Customer can select a delivery slot and proceed to checkout	Pass
PT-09	Payment Method Selection	Selected payment method is accepted successfully	Pass
PT-10	Place Order	Order is created and stored successfully in the database	Pass
PT-11	View Order History	Customer's previous orders are displayed correctly	Pass

PT-12	Admin Product Management	Admin can add, update, view, and delete products successfully	Pass
PT-13	Inventory Management	Product quantity is updated correctly after order processing	Pass
PT-14	Update Order Status	Admin can update order status without errors	Pass
PT-15	Database Connectivity	Application successfully retrieves and stores data in MySQL	Pass

6.2 Test Procedure

The following procedure was used to test the **Grocery Delivery Application**:

1. Start the **Apache Tomcat server** and ensure the MySQL database is running.
2. Launch the **Java Full Stack application** in the web browser.
3. Open the application and verify that the home page and product listings load correctly.
4. Register a new customer account and test the **login authentication**.
5. Browse grocery products and search for different items using the search and category options.
6. View product details and add multiple grocery items to the **shopping cart**.
7. Update product quantities, remove items, and verify that the cart total is calculated correctly.
8. Proceed to checkout, select a **delivery slot**, and choose the preferred payment method.
9. Place the order and verify that the order details are successfully stored in the **MySQL database**.
10. Open the **Admin Dashboard** and test product, category, inventory, customer, and order management.
11. Update the order status and verify that the changes are reflected correctly in the customer's order history.
12. Observe the application's response time, database operations, page loading, and overall performance during normal usage.

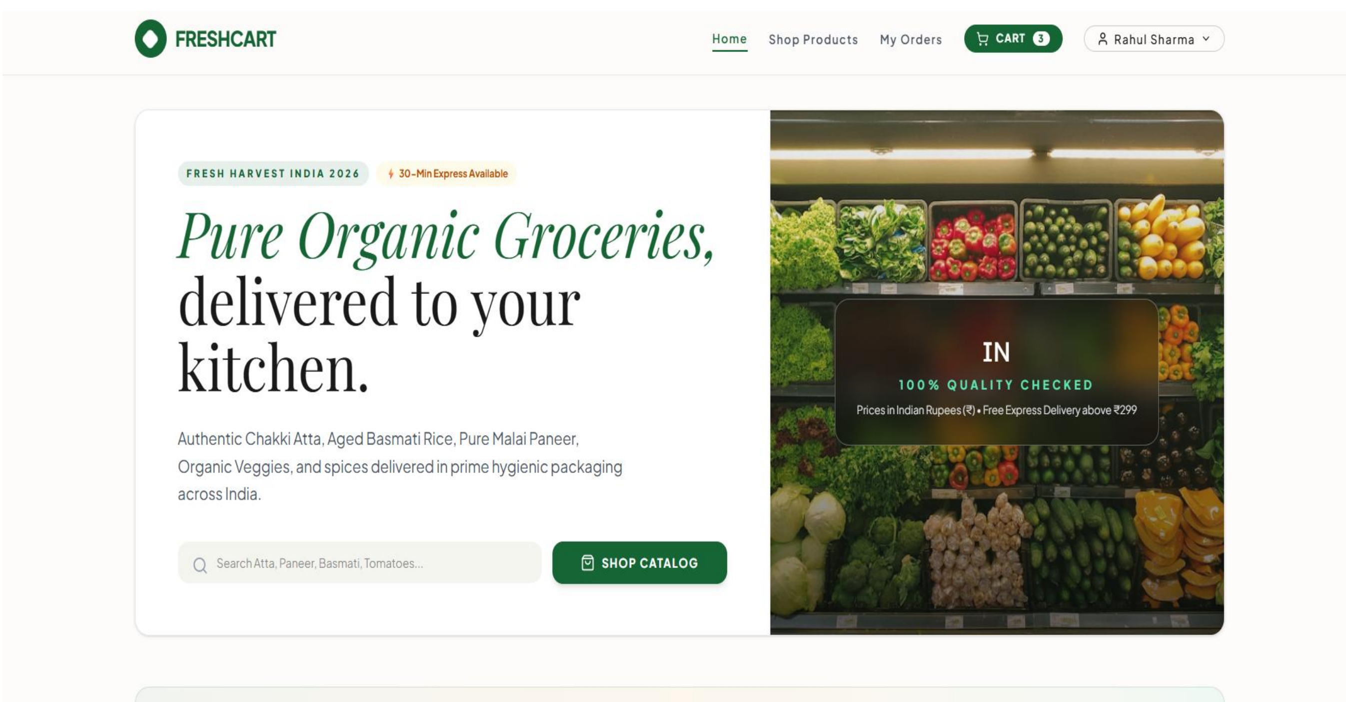
6.3 Performance Outcome

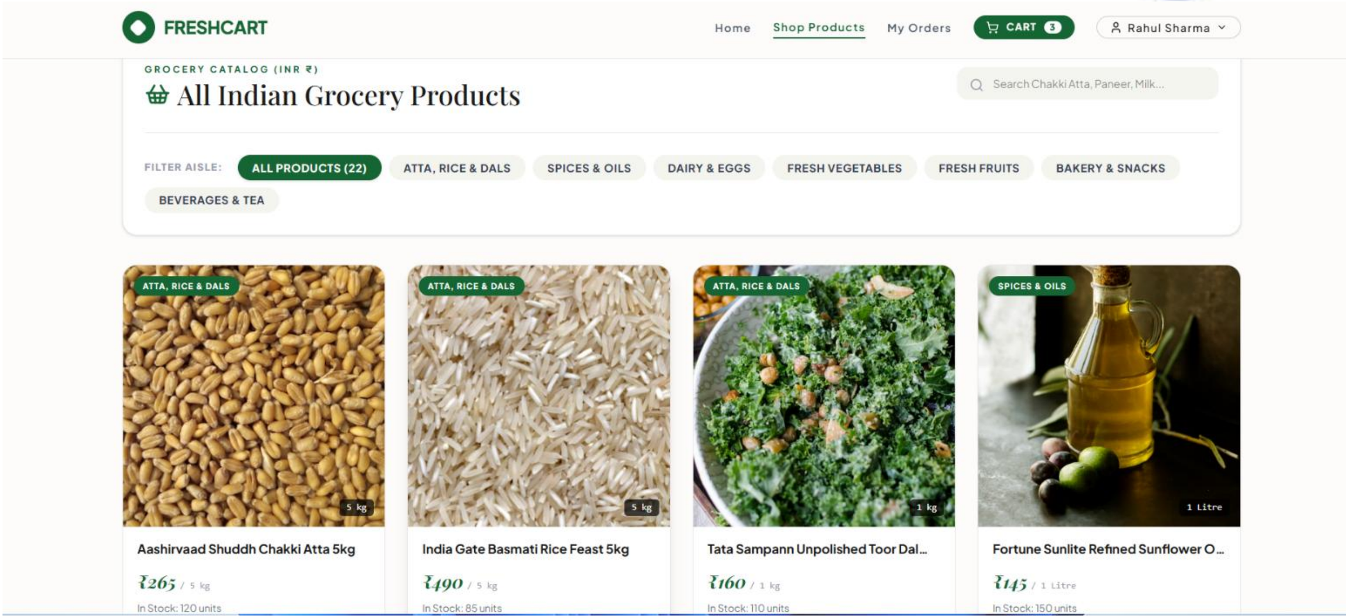
Parameter	Result
Home Page Load Time	~1.5 sec
Product Page Load Time	~1.2 sec
Product Search Response	~200 ms
Database Response Time	~180 ms
Cart Update Response	~150 ms
Order Processing Response	~250 ms

Admin Dashboard Load Time	~1.8 sec
Memory Usage	~100 MB
CPU Usage	Below 35% during normal usage
Concurrent Users Tested	25 Users
Success Rate	100%
Application Availability	Stable

The Grocery Delivery Application demonstrated stable performance under normal operating conditions. The application loaded the main pages efficiently, product search and database operations were completed within acceptable response times, and cart and order processing worked smoothly. The system maintained reasonable CPU and memory usage during normal operations, indicating that the application is suitable for small to medium-scale online grocery delivery services. The modular Java Full Stack architecture also provides scope for future improvements and scalability by enhancing database optimization, server capacity, and deployment infrastructure.

6.4 Project Snapshots




FRESHCART
Home
Shop Products
My Orders
CART 3
Rahul Sharma
GROCERY CATALOG (INR ₹)
All Indian Grocery Products
Search Chakki Atta, Paneer, Milk...
FILTER AISLE:
ALL PRODUCTS (22)
ATTA, RICE & DALS
SPICES & OILS
DAIRY & EGGS
FRESH VEGETABLES
FRESH FRUITS
BAKERY & SNACKS
BEVERAGES & TEA
ATTA, RICE & DALS
5 kg
Aashirvaad Shuddh Chakki Atta 5kg
₹265 / 5 kg
In Stock: 120 units
ATTA, RICE & DALS
5 kg
India Gate Basmati Rice Feast 5kg
₹490 / 5 kg
In Stock: 85 units
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1 kg
Tata Sampann Unpolished Toor Dal...
₹160 / 1 kg
In Stock: 110 units
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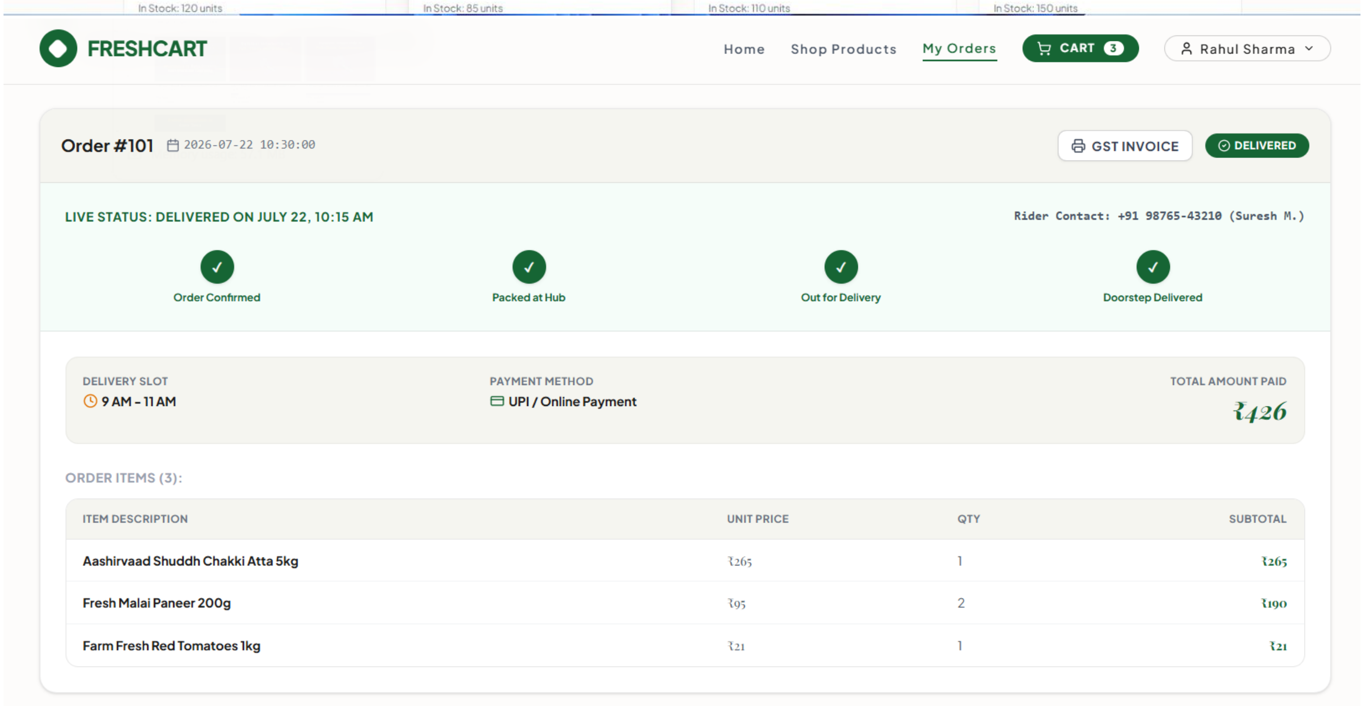
1 Litre
Fortune Sunlite Refined Sunflower O...
₹145 / 1 Litre
In Stock: 150 units


FRESHCART
Home
Shop Products
My Orders
CART 3
Rahul Sharma
Order #101
2026-07-22 10:30:00
GST INVOICE
DELIVERED
LIVE STATUS: DELIVERED ON JULY 22, 10:15 AM
Rider Contact: +91 98765-43210 (Suresh M.)
Order Confirmed
Packed at Hub
Out for Delivery
Doorstep Delivered
DELIVERY SLOT
9 AM - 11 AM
PAYMENT METHOD
UPI / Online Payment
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ORDER ITEMS (3):
ITEM DESCRIPTION
UNIT PRICE
QTY
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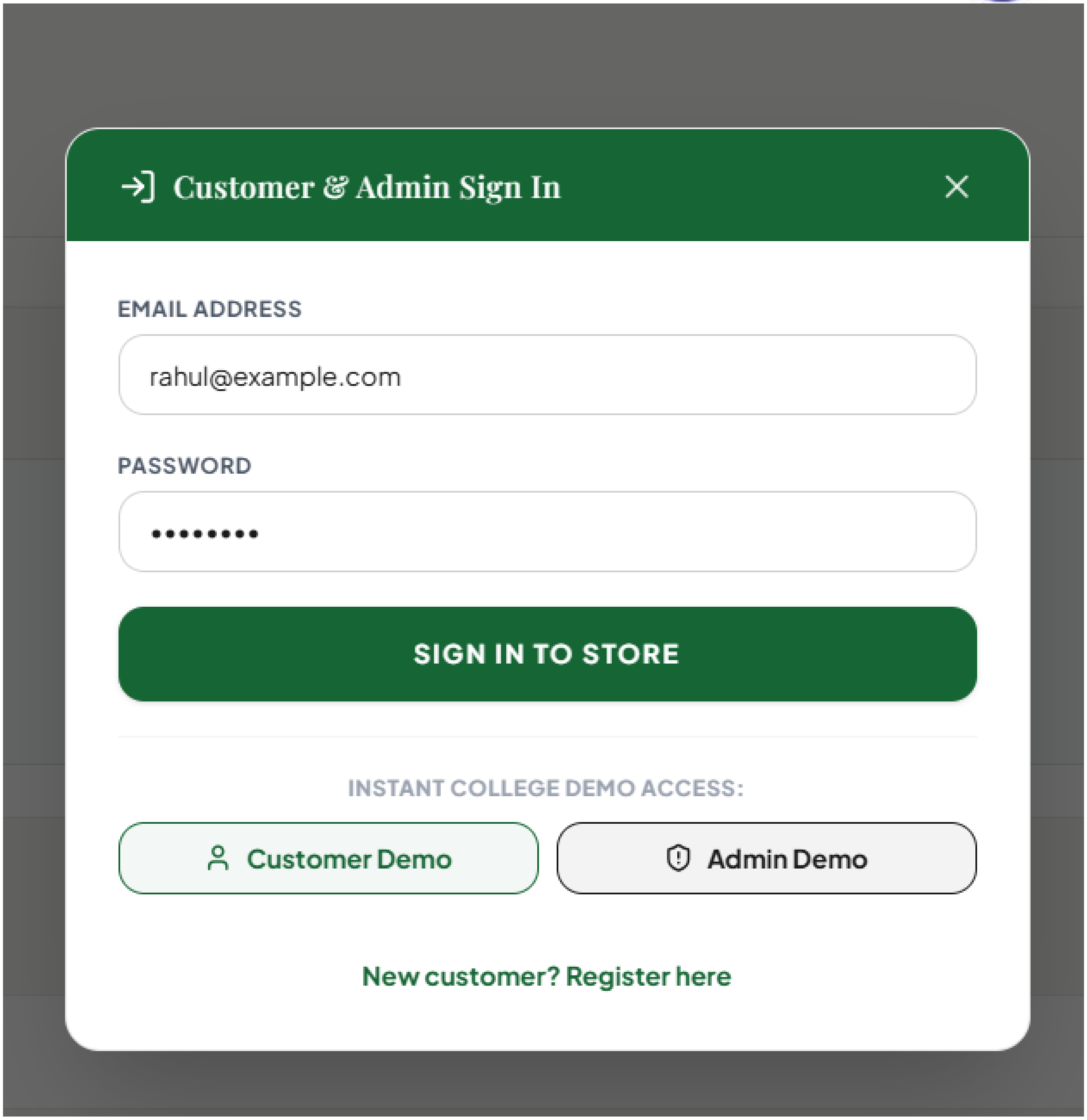
Customer & Admin Sign In
EMAIL ADDRESS
rahul@example.com
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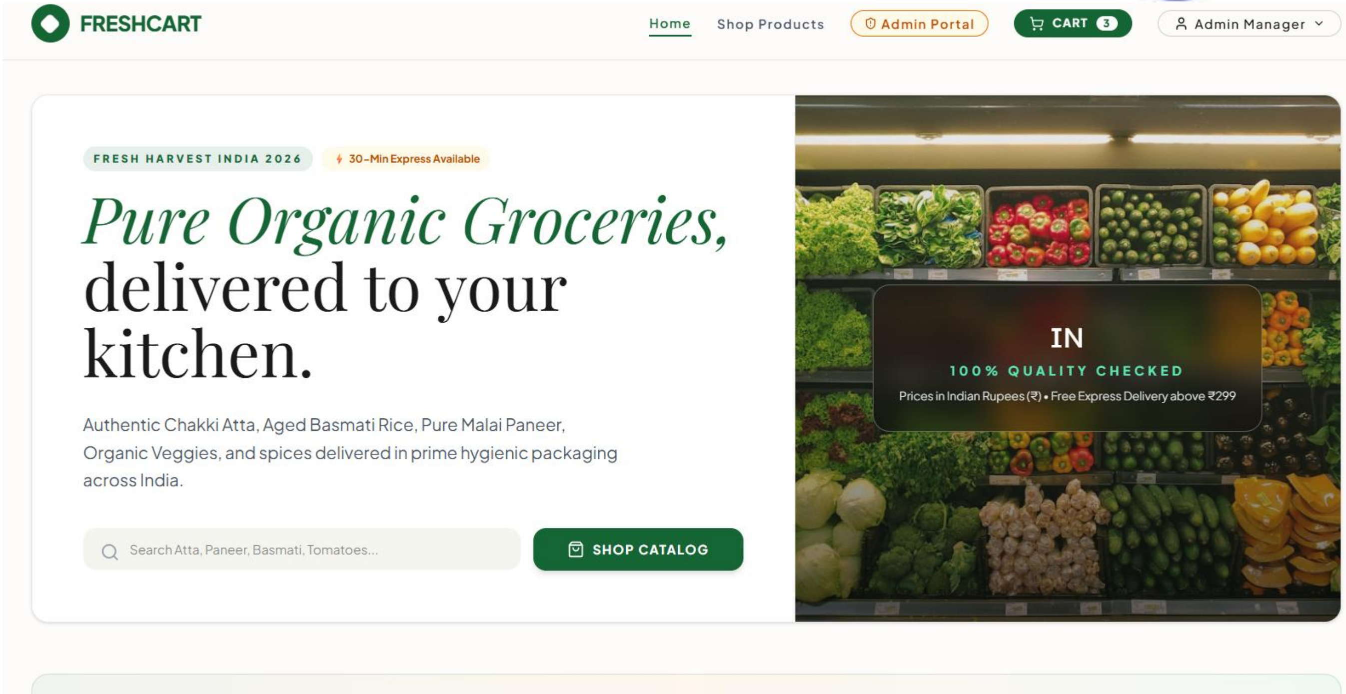
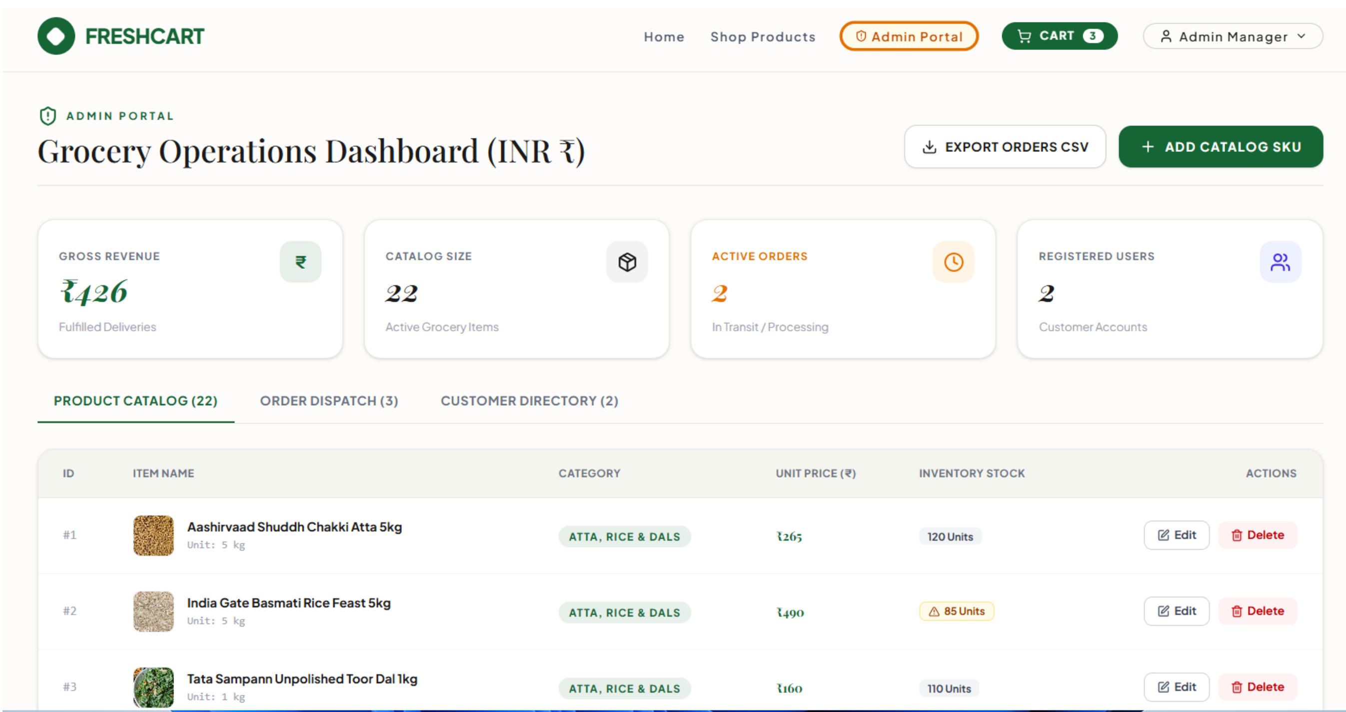
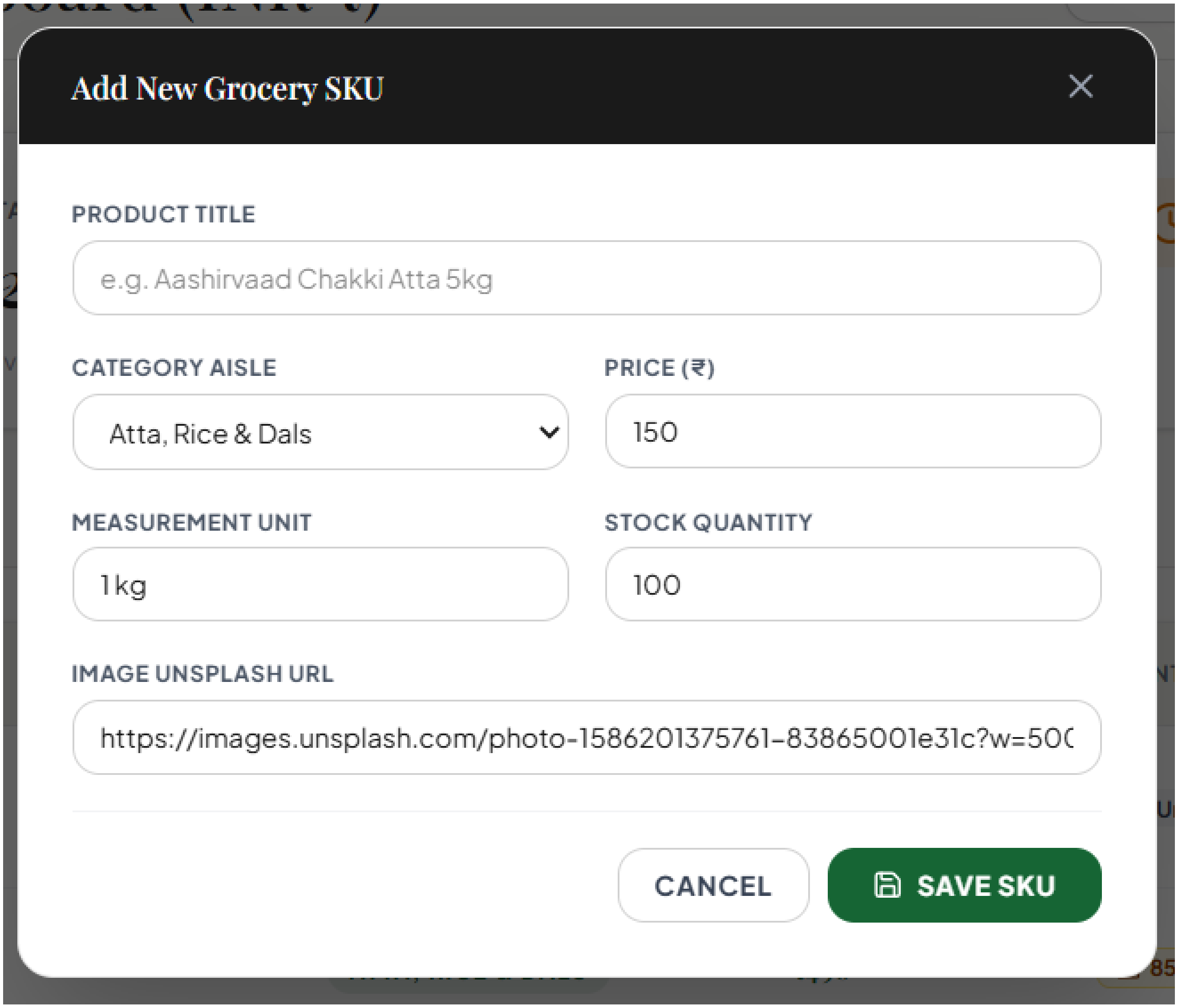
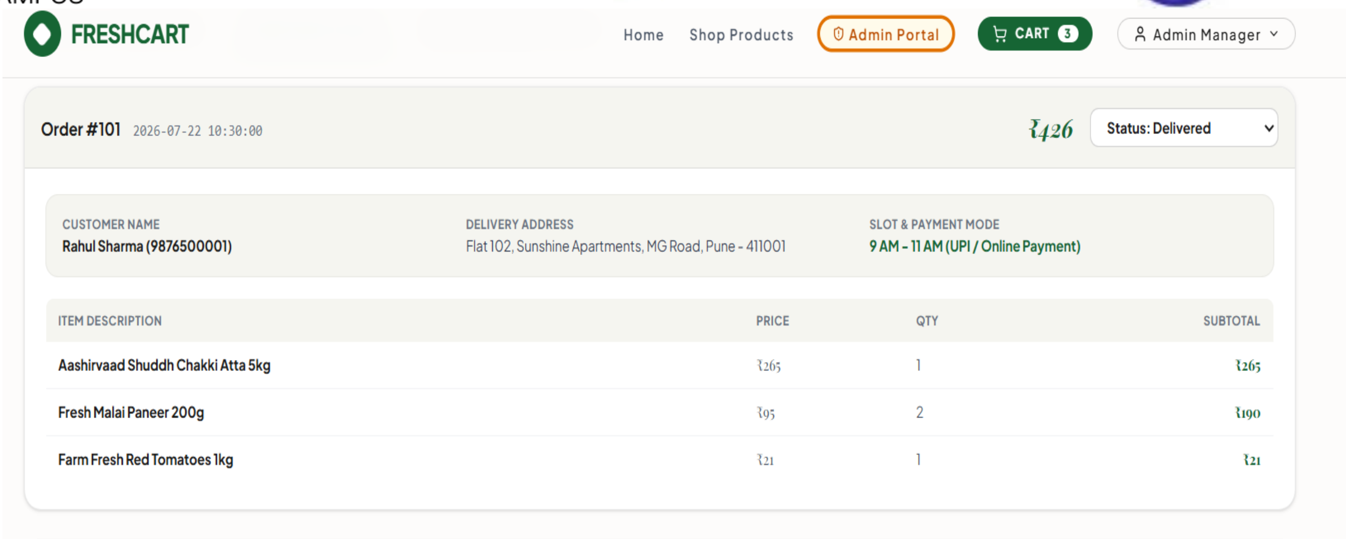
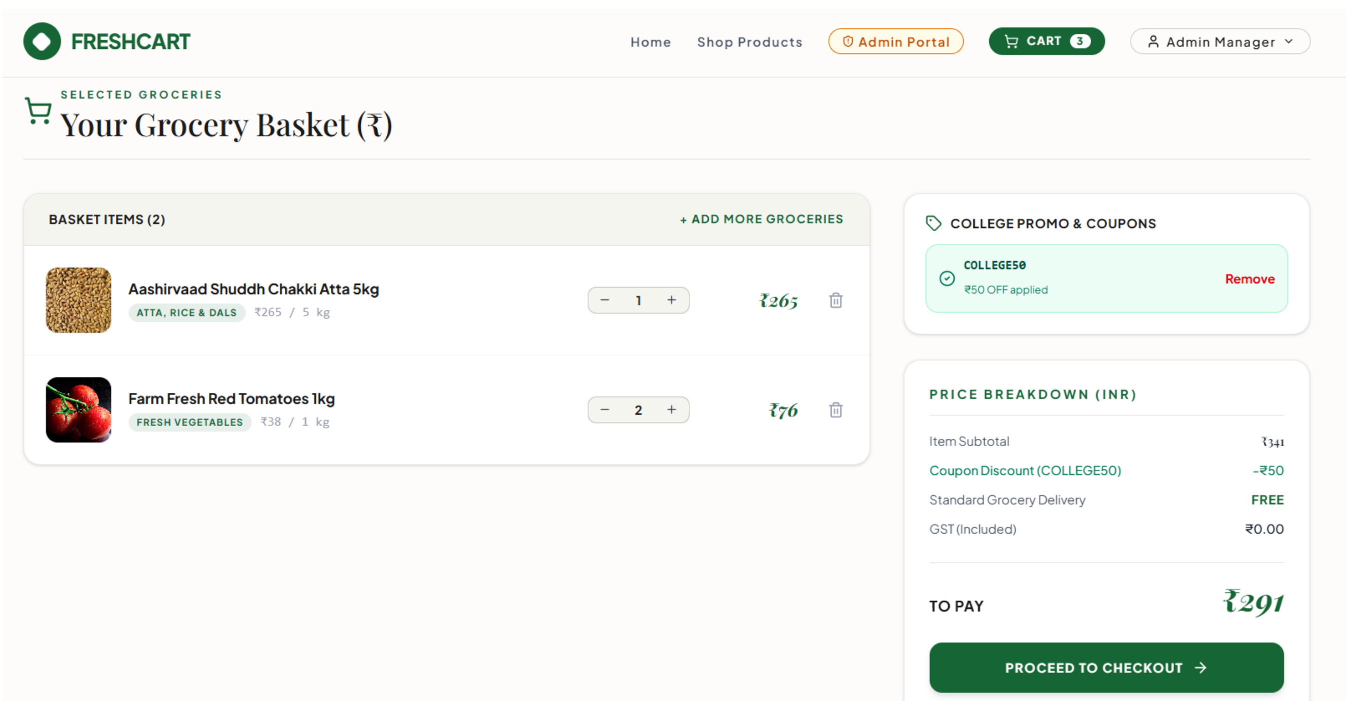
SIGN IN TO STORE
INSTANT COLLEGE DEMO ACCESS:
Customer Demo
Admin Demo
New customer? Register here


FRESHCART
Home
Shop Products
Admin Portal
CART 3
Admin Manager
FRESH HARVEST INDIA 2026
30-Min Express Available
Pure Organic Groceries,
delivered to your kitchen.
Authentic Chakki Atta, Aged Basmati Rice, Pure Malai Paneer, Organic Veggies, and spices delivered in prime hygienic packaging across India.
Search Atta, Paneer, Basmati, Tomatoes...
SHOP CATALOG
IN
100% QUALITY CHECKED
Prices in Indian Rupees (₹) • Free Express Delivery above ₹299


FRESHCART
Home
Shop Products
Admin Portal
CART 3
Admin Manager
ADMIN PORTAL
Grocery Operations Dashboard (INR ₹)
EXPORT ORDERS CSV
ADD CATALOG SKU
GROSS REVENUE
₹426
Fulfilled Deliveries
CATALOG SIZE
22
Active Grocery Items
ACTIVE ORDERS
2
In Transit / Processing
REGISTERED USERS
2
Customer Accounts
PRODUCT CATALOG (22)
ORDER DISPATCH (3)
CUSTOMER DIRECTORY (2)
ID
ITEM NAME
CATEGORY
UNIT PRICE (₹)
INVENTORY STOCK
ACTIONS
#1
Aashirvaad Shuddh Chakki Atta 5kg
Unit: 5 kg
ATTA, RICE & DALS
₹265
120 Units
Edit
Delete
#2
India Gate Basmati Rice Feast 5kg
Unit: 5 kg
ATTA, RICE & DALS
₹490
85 Units
Edit
Delete
#3
Tata Sampann Unpolished Toor Dal 1kg
Unit: 1 kg
ATTA, RICE & DALS
₹160
110 Units
Edit
Delete


Add New Grocery SKU
PRODUCT TITLE
e.g. Aashirvaad Chakki Atta 5kg
CATEGORY AISLE
Atta, Rice & Dals
PRICE (₹)
150
MEASUREMENT UNIT
1kg
STOCK QUANTITY
100
IMAGE UNSPLASH URL
https://images.unsplash.com/photo-1586201375761-83865001e31c?w=50(
CANCEL
SAVE SKU


FRESHCART
Home
Shop Products
Admin Portal
CART 3
Admin Manager
Order #101 2026-07-22 10:30:00
₹426
Status: Delivered
CUSTOMER NAME
Rahul Sharma (9876500001)
DELIVERY ADDRESS
Flat 102, Sunshine Apartments, MG Road, Pune - 411001
SLOT & PAYMENT MODE
9 AM - 11 AM (UPI / Online Payment)
ITEM DESCRIPTION
PRICE
QTY
SUBTOTAL
Aashirvaad Shuddh Chakki Atta 5kg ₹265 1 ₹265
Fresh Malai Paneer 200g ₹95 2 ₹190
Farm Fresh Red Tomatoes 1kg ₹21 1 ₹21


FRESHCART
Home
Shop Products
Admin Portal
CART 3
Admin Manager
SELECTED GROCERIES
Your Grocery Basket (₹)
BASKET ITEMS (2)
+ ADD MORE GROCERIES
Aashirvaad Shuddh Chakki Atta 5kg
ATTA, RICE & DALS ₹265 / 5 kg
1
₹265
Farm Fresh Red Tomatoes 1kg
FRESH VEGETABLES ₹38 / 1 kg
2
₹76
COLLEGE PROMO & COUPONS
COLLEGE50
₹50 OFF applied
Remove
PRICE BREAKDOWN (INR)
Item Subtotal ₹341
Coupon Discount (COLLEGE50) -₹50
Standard Grocery Delivery FREE
GST (Included) ₹0.00
TO PAY ₹291
PROCEED TO CHECKOUT

7. My Learnings

The development of the Grocery Delivery Application provided me with valuable hands-on experience in designing and developing a Java Full Stack web application. Throughout the project, I enhanced my knowledge of Java, JSP, Servlets, JDBC, MySQL, HTML, CSS, Bootstrap, and JavaScript, while understanding how frontend, backend, and database components work together to create a complete web application.

I learned how to design a structured application architecture, implement CRUD operations, establish database connectivity using JDBC, and develop separate functionalities for customers and administrators. Working on features such as product management, category management, shopping cart operations, delivery slot selection, payment method selection, order processing, inventory management, and order history improved my problem-solving and logical thinking skills.

This project also strengthened my understanding of software development practices such as responsive UI design, form validation, session management, database operations, debugging, testing, exception handling, and performance optimization. I gained practical experience in designing database tables and establishing relationships between users, products, carts, orders, and order items.

Overall, this project significantly improved my technical knowledge, coding practices, problem-solving abilities, and understanding of full-stack application development. The experience increased my confidence in developing Java-based web applications and provided a strong foundation for future opportunities in software development and full-stack web development.

8. Future Work Scope

The Grocery Delivery Application has been developed as a functional Java Full Stack online grocery shopping platform. However, several enhancements can be implemented in the future to improve its functionality, scalability, security, and user experience.

Future improvements can include integrating a real online payment gateway such as Razorpay, Stripe, or another suitable payment service instead of the current demo payment functionality. The delivery module can also be enhanced with real-time order tracking and map integration to allow customers to monitor their delivery status and estimated arrival time.

The application can be improved by implementing advanced product recommendations based on customer purchase history, frequently purchased products, and product preferences. Features such as wishlist management, favourite products, product ratings and reviews, discount coupons, loyalty and reward programs, and personalized offers can also be added.

For better inventory management, the system can include low-stock notifications, automatic stock updates, sales reports, and inventory analytics. Administrators can also be provided with advanced dashboards displaying order statistics, revenue information, popular products, and customer activity.

The security of the application can be further improved through stronger password protection, role-based authorization, secure session management, and enhanced input validation. The application can also be deployed on cloud platforms such as AWS, Microsoft Azure, or Google Cloud to improve availability and scalability.

Additional features such as email/SMS notifications, multiple delivery addresses, multilingual support, mobile application support, subscription-based grocery delivery, and automatic repeat ordering can further enhance the system.

Overall, these future enhancements would make the Grocery Delivery Application more secure, scalable, convenient, and suitable for real-world commercial deployment.